Bridging the Educational Gap in Robotics: A Hands-on Tutorial Framework for Reinforcement Learning

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INTRODUCTION

State-of-the-art topics in robotics, such as deep Reinforcement Learning, are critical for modern applications, but are often perceived as very advanced for higher-education students. The situation hinders the substantial integration of such topics into the syllabus, creating a knowledge gap. Subjects like Proximal Policy Optimization and Sim2Real are typically confined to graduate-level study. We developed an educational framework presented as a 6-hour hands-on tutorial. It was first delivered at LARS 2025, and used advanced simulation tools such as NVIDIA IsaacLab to make these challenging, researchlevel concepts accessible and engaging for a broader audience.

PROPOSED METHODOLOGY

The tutorials were structured as modules that introduced RL concepts and their applications to robotics, following recommendations from recent literature on RL education [3]. This initiative was implemented as a six-hour tutorial during the 2025 Latin American Robotics Symposium (LARS). Through short lectures and hands-on tutorials using Python, participants engaged with topics ranging from fundamental RL theory to advanced robotic control using NVIDIA Isaac Sim and Isaac Lab. Topics included neural networks, Markov Decision Processes, Monte Carlo method, Temporal Difference learning, Q-Learning, SARSA and Deep Q-Learning

The second part focused on implementing RL algorithms for robotic control. Participants explored PPO within Isaac Sim and Isaac Lab, applying them to simulated robotic arms and quadruped locomotion tasks. In addition, a Sim2Real demo was deployed onto a Unitree Go2 quadruped robot. Supplementary material for the tutorials was gathered from official online documentation and open repositories [1][4], including the PyTorch and CleanRL frameworks, OpenSource AI educational resources, and Unitree's SDK2 [5], MuJoCo, and RL Lab platforms. All adapted examples and supporting code are available at https://github.com/EMendezJ/LARS-RL-TUTORIAL.

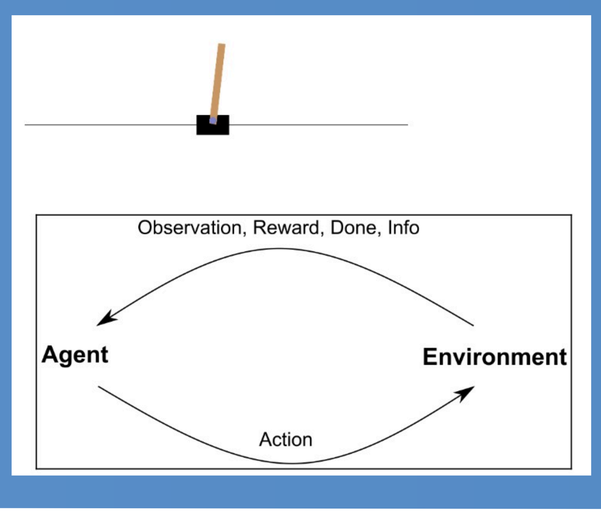


Fig. 1. Initial benchmarking of policy-gradient methods using the CartPole-v1 control environment for fundamental stability analysis.

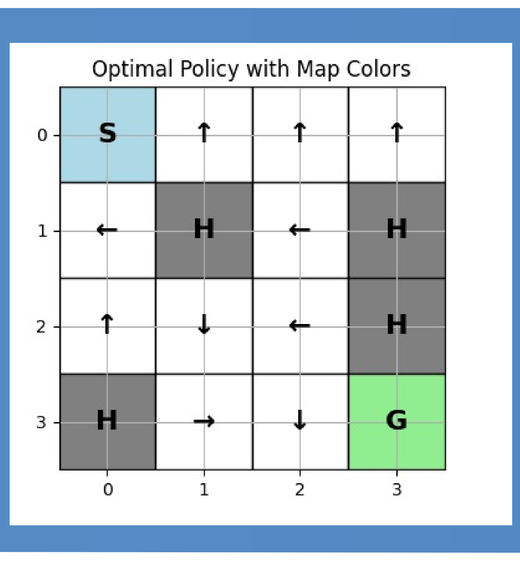


Fig. 2. Discrete state-space navigation using a Markov Decision Process (MDP) framework within the FrozenLake stochastic environment.

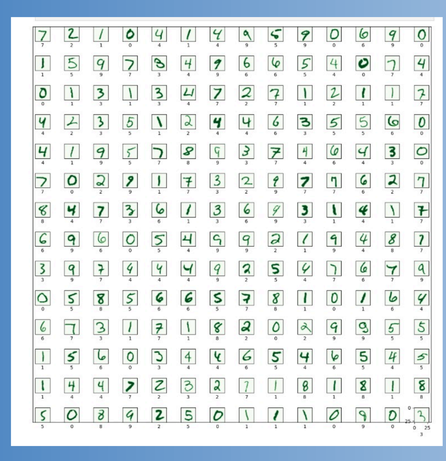


Fig. 3. Convolutional Neural Network (CNN) performance for high-accuracy classification and pattern recognition in supervised learning benchmarks.

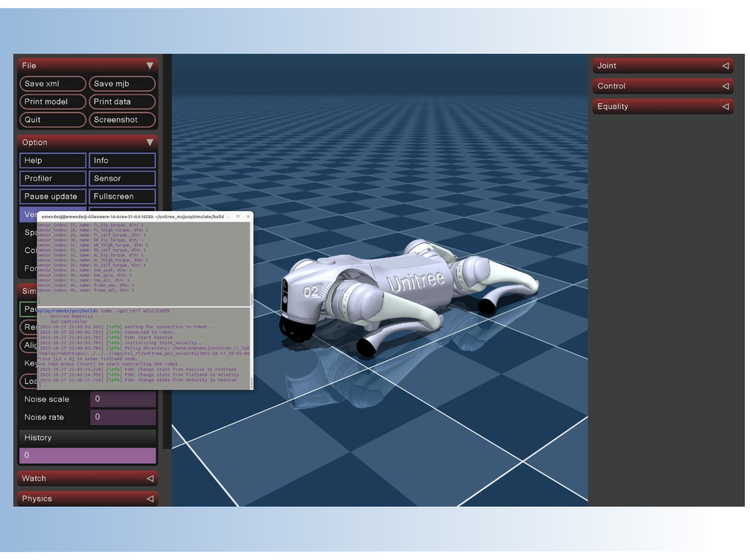


Fig. 4. High-fidelity MuJoCo physics engine integration for Sim2Sim validation of reinforcement learning policies prior to hardware deployment.

RESULTS

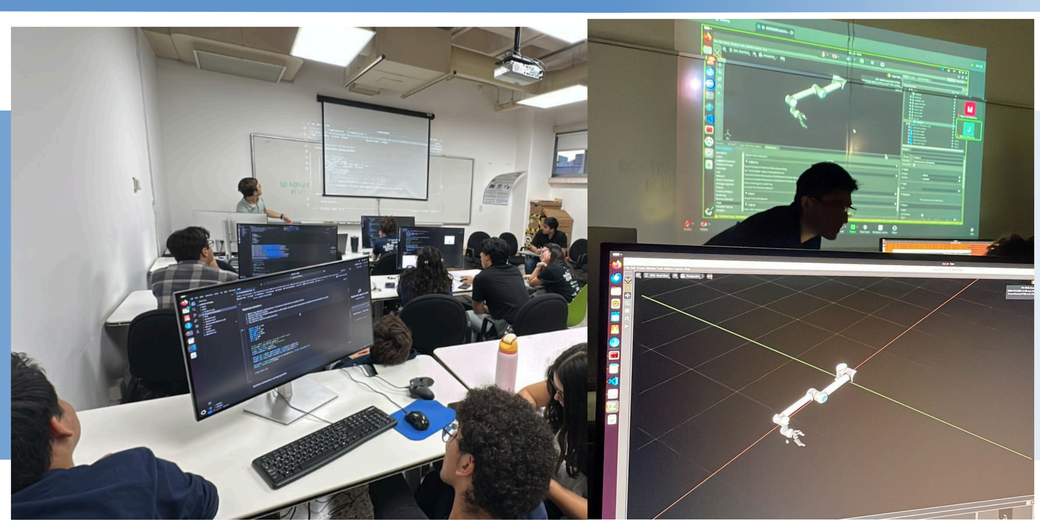


Fig. 5. Experimental validation and real-time interaction during LARS 2025.



Fig. 6. Sim2Real locomotion policy deployment during LARS2025.

The framework was validated at the Latin American Robotics Symposium (LARS) 2025.

- **Simulation Performance:** Participants learned fundamental Reinforcement Learning topics, implemented PPO for robotic arm control and quadruped locomotion tasks within the Isaac environment.
- **Hardware Validation:** A successful Sim2Real demo was executed using a Unitree Go2 quadruped robot.
- **Educational Impact:** The results demonstrate that structured, simulation-driven learning experiences effectively deliver research-level topics to a broad academic audience.

CONCLUSION

This work presented a hands-on educational initiative that bridges the gap between RL theory and robotic applications through an accessible tutorial format. By combining interactive Python environments, simulation platforms, and real-robot demonstrations, the tutorials effectively engaged participants in understanding RL concepts and their relevance to robotics. Future work will focus on expanding and refining the GitHub repository into a self-contained online learning platform, featuring updated examples, clear documentation, and interactive resources that allow users to follow the material independently, without the need for a live tutorial.

References

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[4] M. Jamali, "Reach: Repository for reachability analysis in robotics," https://github.com/mobinajamali/Reach, 2023, accessed: Nov. 5, 2025.

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